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# Explainability in Reservoir Well-logging Evaluation: Comparison of Variable Importance Analysis with Shapley Value Regression, SHAP and LIME

**Shaogui Deng[1]. Chris Aldrich.[2] Xiu Liu.[3] Fengjiao Zhang[1, 4]**

*[1]National Key Laboratory of Deep Oil and Gas, China University of Petroleum (East China), Qingdao 266580, P. R. China (e-mail: dengshg@upc.edu.cn).*

*[2]Western Australian School of Mines: Minerals, Energy and Chemical Engineering, Curtin University, GPO Box U1987, Perth, WA, 6845, Australia (e-mail: chris.aldrich@curtin.edu.au)*

*[3]Western Australian School of Mines: Minerals, Energy and Chemical Engineering, Curtin University, GPO Box U1987, Perth, WA, 6845, Australia (e-mail: xiu.liu@curtin.edu.au)*

*[4]Western Australian School of Mines: Minerals, Energy and Chemical Engineering, Curtin University, GPO Box U1987, Perth, WA, 6845, Australia (e-mail: 1967468598@qq.com)*

**Abstract**: Machine learning algorithms have made significant progress in the logging evaluation field, but their "black box" characteristics make is a hindrance to the interpretation and acceptance of these models. In the face of a multitude of interpretation methods, choosing the most suitable becomes a challenge. In this study, a random forest regression model was first applied to a simulated dataset. Subsequently, Shapley value regression, two variants of Shapley Additive Explanation (SHAP), i.e. KernelSHAP and TreeSHAP, and Local Interpretable Model-Agnostic Explanation (LIME) methods were compared for global and local explanation of the random forest model. In addition, a comparative analysis of these explanatory methods was carried out using shear wave velocity prediction as an example. It was found that all four methods provided an accurate analysis of the variable importance rankings. Regardless of the global or local interpretation, the Shapley value method gave the most reliable results, closely followed by the KernelSHAP method. In addition, it was observed that LIME outperformed TreeSHAP in terms of global interpretation, but not necessarily in terms of local explanation.



*Keywords*: Machine learning; Shapley value; SHAP; LIME; Well logging.

## 1. INTRODUCTION

Machine learning models, with their superior data processing and analytical capabilities, have achieved remarkable success in several fields, such as oil and gas exploration, healthcare, finance, self-driving cars, speech recognition, image processing, and recommender systems (Aghamohammadi et al., 2019; Amann et al., 2020; Ni et al., 2020; Zhao et al., 2022).

However, most machine learning algorithms are so-called "black box" models, since the way that they arrive at decisions is generally difficult to explain to humans. This reduces the credibility of the models and renders their deployment in mission critical applications less appealing (Ribeiro et al., 2016; Rodríguez-Pérez & Bajorath, 2020). Therefore, improving the transparency and interpretability of machine learning algorithms has become an essential focus of current research to ensure the rationality of decision-making and to promote the models' continuous optimization.

Currently, there is a rapid development of methods in the field of machine learning that are able to provide local or global explanations by developing explanatory models to elucidate the model's operation and decision-making process after model training is completed. In this context, local explanations refer to the explanation of specific instances or samples, while global explanation refers to the overall importance of a specific variable or predictor in the model. These methods include among other permutation feature importance, partial dependence plots, local interpretable model-agnostic explanations (LIME) and Shapley additive explanations (SHAP) (Breiman, 2001; Friedman, 2001; Lundberg & Lee, 2017; Ribeiro et al., 2016).

Interpretable or explanatory methods have been applied in the field of reservoir logging evaluation, such as porosity prediction, lithofacies classification, capacity prediction, and permeability prediction (Chen et al., 2023; Ma et al., 2023; Voskresenskiy et al., 2021).

However, despite the diversity of these diagnostic methods, different researchers have different definitions of reliable explanation, which means that there is strong subjectivity in measuring the quality of explainability. As a result, the lack of metrics to measure the quality of interpretive results makes it difficult for scholars to choose the one that best suits the research question among the many interpretive approaches (Fries & Ryden, 2022).

Therefore, the most popular full Shapley, SHAP and LIME methods are compared in this study. Firstly, a simulation example is used to assess the performance of different interpretation methods in global and local interpretation using


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correlation coefficients as an evaluation criterion. Then an example from reservoir logging evaluation, shear wave velocity prediction, is used to validate the results.

## 2. METHODS

### *2.1 Random Forests*

The Random Forest algorithm was chosen as the predictive model for this study, since it can provide highly accurate predictions and is generally resistant to overfitting. When comparing the performance of the explanatory algorithms, the model explained is the RF model.

RF is an integrated learning algorithm that improves accuracy by constructing multiple decision trees and aggregating their predictions. It uses a self-sampling method to randomly select samples from the training data, while randomly selecting features during the construction of each tree to increase the diversity of the model. With classification task, random forests make the final prediction based on majority voting. With regression tasks, the average of the predicted values of all trees is calculated.

### *2.2 Shapley Value methods*

Shapley value methods or Full Shapley methods as referred to here are used to measure the marginal contribution of each feature to the prediction result. For feature $j$, the marginal contribution of the $j$-th feature is computed for all possible coalitions $S$ (containing participants other than $j$) by considering the values change after adding the $j$-th feature to $S$.

For a dataset containing M features, the formula for calculating the Shapley value ($\phi_j(v, M)$) of the $j$-th participant is shown in (1).

$$\phi_j(v, M) = \sum_{S \subseteq M \setminus \{j\}}^{M} \frac{|S|!\,(M - |S| - 1)!}{M!} (v(S \cup \{j\}) - v(S)) \tag{1}$$

Where, $M \setminus \{j\}$ denotes all participants after excluding the $j$-th participant; in weight $\frac{|S|!(M-|S|-1)!}{M!}$, $M!$ is the number of all combinations of M features, $|S|!\,(M - |S| - 1)!$ is the number of all possible combinations after removing the $j$-th feature; and $v(S)$ denotes the predicted output corresponding to the feature combination $S$.

### *2.3 Shapley Additive Explanations (SHAP)*

Since all possible combinations of features need to be considered for the accurate calculation of the Shapley value, the number of combinations grows exponentially when the number of features increases, eventually resulting in a prohibitively costly calculation process.

To overcome this challenge, various techniques have been developed to simplify the computation of Shapley values, the most popular of which is SHAP. It not only retains the theoretical advantages of the Shapley value method, but also makes it more feasible for practical applications. To main variants are considered here: KernelSHAP and TreeSHAP.

*KernelSHAP*: This method employs a kernel-based approach to estimating Shapley values. KernelSHAP is applicable to various types of models, including linear, nonlinear, and mixed models. It approximates the feature contributions by constructing a weighted least squares regression problem with weights determined by the presence or absence of features. This method, while more general, may also be computationally expensive with a large number of features.

*TreeSHAP*: It is specially designed for tree-based models (such as decision trees, random forests, gradient boosters). It exploits the properties of tree structures to optimize the computational process, making it more efficient to compute Shapley values. TreeSHAP computes the expected values of feature contributions directly on the tree model by taking into account the specific order of tree paths and tree node splits. This approach is particularly effective when dealing with large tree models, as it significantly reduces the computational complexity.

### *2.4 Local Interpretable Model-Agnostic Explanations (LIME)*

The key to LIME is that it creates a simple local model (often a linear regression model or a decision tree) to approximate the behavior of a complex model near a specific prediction point. LIME reveals how the model makes decisions in a localized region by analyzing the target model's response to slightly perturbed samples. The advantage of this local approximation approach is its generality to various types of machine learning models, providing intuitive explanations of model predictions.

### *2.5 Evaluation metrics*

There is no recognized objective mathematical criterion for evaluating the validity of the interpretation given its subjective nature. In this study, we adopt the correlation coefficient RI as a measure to assess the accuracy of the various interpretation methods (2). It is worth noting that an explanatory model cannot be evaluated by the error of the calculated coefficients in relation to the significance calculated by the explanatory method, since only the Full Shapley and KernelSHAP methods have the same computational principle.

$$\mathrm{RI} = \frac{cov(Y, P)}{\sqrt{D(Y)D(P)}} \tag{2}$$

In (2), $cov(Y, P)$ represents the covariance function that characterizes the interrelationship between the actual value $Y$ and the predicted value $P$. The larger the RI value, the better the performance of the interpretation method.

## 3. CASE STUDY: SIMULATED DATA

### *3.1 Data analysis*

To evaluate the effectiveness of various model interpretation methods, a simulated dataset was used to test the reliability of the Shapley method in explaining model behaviors. This dataset, comprising 1000 samples, was generated based on a multivariate normal distribution and encompasses five features ($\mathbf{X} = [x_1, x_2, x_3, x_4, x_5]$), with a mean vector of [0 0 0 0 0] and a covariance matrix as in (3). The target variable y is generated by dot-multiplying the data X with a vector of coefficients, which are [2.6, 2, 1.5, 1, 0.1].

$$\sum = \begin{bmatrix} 1 & 0.3 & 0.3 & 0.3 & 0.3 \\ 0.3 & 1 & 0.3 & 0.3 & 0.3 \\ 0.3 & 0.3 & 1 & 0.3 & 0.3 \\ 0.3 & 0.3 & 0.3 & 1 & 0.3 \\ 0.3 & 0.3 & 0.3 & 0.3 & 1 \end{bmatrix} \tag{3}$$

With reference to (2), the actual values were not those of the target value y, but the coefficients of the matrix, i.e. $Y$ = [2.6, 2, 1.5, 1, 0.1] associated with the five variables, while the predicted values $P$ are the coefficient values calculated by each explanation model.

*3.2 RF for regression*

The dataset was randomly split into training and testing subsets following an 8:2 ratio. Subsequently, a Random Forest algorithm was employed to model the data. In this setup, each split node in the decision trees considered a maximum of five features. The model was configured with optimal hyperparameters: The total number of trees in the forest (K) was set to 100, and the minimum number of samples required at each leaf node ($n_{try}$) was established at 5. As expected with a linear relationship, the model demonstrates high predictive accuracy on the test set, maintaining a stable $R^2$ value of approximately 0.98.

*3.3 Explanation methods comparison*

(1) Global explanations

The simulated linear data have simplicity and transparency, and the magnitudes of the coefficients indicate the effect on the target variable when the feature is changed by one unit. Thus, the importance of each feature is proportional to the absolute magnitude of its coefficient, i.e. the actual ranking of importance is $x_1 > x_2 > x_3 > x_4 > x_5$.

The performance of the Full Shapley, TreeSHAP, KernelSHAP and LIME methods in terms of global interpretation can consequently be considered. Fig. 1 depicts the feature importance values (model coefficients) derived from 20 runs on the test dataset, highlighting the results obtained with Shapley in Fig. 1(a), TreeSHAP in Fig. 1(b), KernelSHAP in Fig. 1(c), and LIME values in Fig. 1(d). Within each "violin," a white dot illustrates the median value, while the internal thick bars denote the interquartile range (IQR), ranging from the 25th percentile (Q1) to the 75th percentile (Q3). The contours of each violin sketch the data's distribution; where the violin broadens, it indicates a higher density of data points. In contrast, the narrow tips suggest sparser regions, aligning with the tails of the distribution. Data points that lie outside the IQR yet within the extended lines, or "whiskers," signify the range up to 1.5 times the IQR, and dots beyond this expanse may be considered outliers.

As shown in Fig. 1, the global importance values generated by the four interpretation methods are consistent with the actual values, demonstrating their efficacy in clarifying the importance of variables in the model.

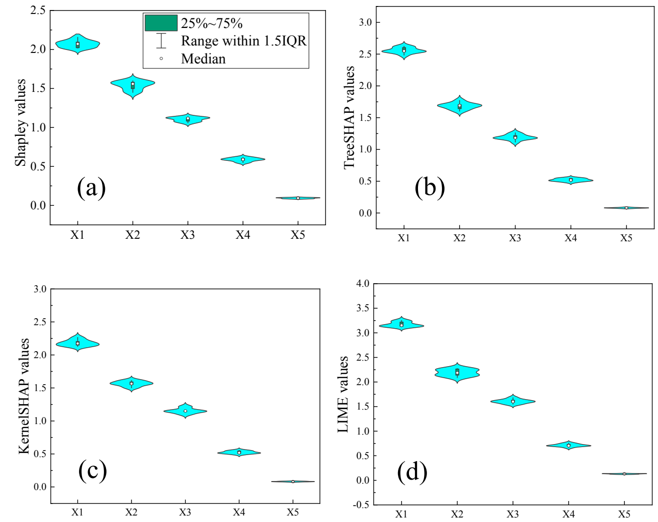


Figure 1. Ranking of variables importance. (a) Full Shapley, (b) TreeSHAP, (c) KernelSHAP and (d) LIME.

In order to further compare and visualize the accuracy of the four interpretation methods, the results of one randomly selected trial were used to produce a rendezvous plot of interpreted importance versus matrix coefficients, also known as actual importance (Fig. 2). It shows that the Full Shapley value has the best correlation with the coefficients, and the $RI^2$ can reach 0.98.

Further, Fig. 3 illustrates the $RI^2$ distribution for the 20 trials. It can be seen that the Shapley values are closest to the actual scenario, followed by the KernelSHAP, LIME and TreeSHAP methods. The above results demonstrate the superiority of the game theory-based approach in model interpretation. In addition, as an approximation of the Shapley values, KernelSHAP is more accurate than TreeSHAP, but the TreeSHAP method is more computationally efficient for tree models only.

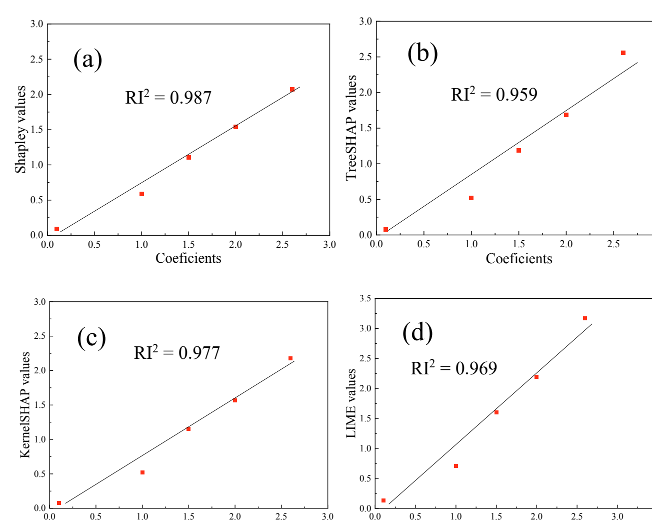


Figure 2. The rendezvous plots of real and interpreted importance. (a) Full Shapley, (b) TreeSHAP, (c) KernelSHAP and (d) LIME.

Therefore, for global interpretation, the Shapley values best reflect the true importance distribution of the variables in terms of interpretation accuracy, followed by the KernelSHAP, LIME and TreeSHAP methods.

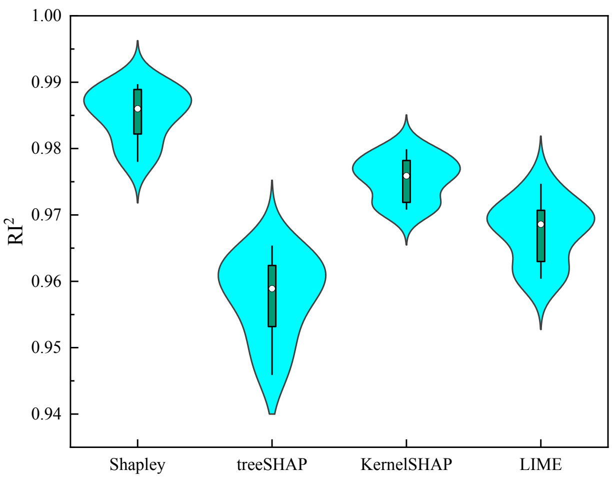


Figure 3. $RI^2$ distributions for 20 trials of the four interpretation methods.

(2) Local explanations

In terms of local interpretation of individual data, the contribution of each feature $x_i$ to the model prediction can be quantified as the product of its coefficient and the feature value in the context of a linear relationship.

Twenty points were randomly selected to produce the $RI^2$ value distribution (Fig. 4). It can be seen that the Shapley values have the most concentrated distribution of $RI^2$ values with the matrix coefficients, followed by the KernelSHAP, TreeSHAP and LIME values.

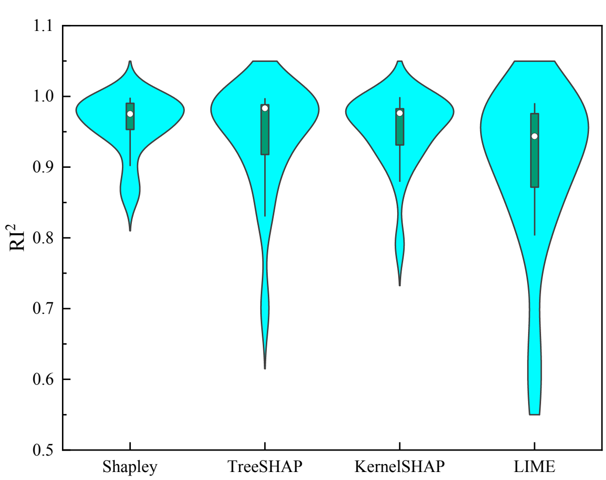


Figure 4. $RI^2$ distributions for 20 samples of the four interpretation methods.

By further analyzing the characteristics of the results of each interpretation method as the variables $x_1$, $x_2$, $x_3$, and $x_4$ change, it was found that the distribution of Shapley values was closest to that of actual values, followed by KernelSHAP values, TreeSHAP values and LIME values (Fig. 5). Moreover, the stronger the relationship between the **x** values and the y value, the closer the calculated importance value is to the true value.

The results of the LIME method exhibit the characteristics of clustering, which means that the LIME values are highly similar when the variables are within a certain range. The stronger the correlation between the variables and the y-values, the more obvious this feature is. The conjecture is that this is because the LIME method uses the same linear model for approximating the data in certain ranges in its calculations.

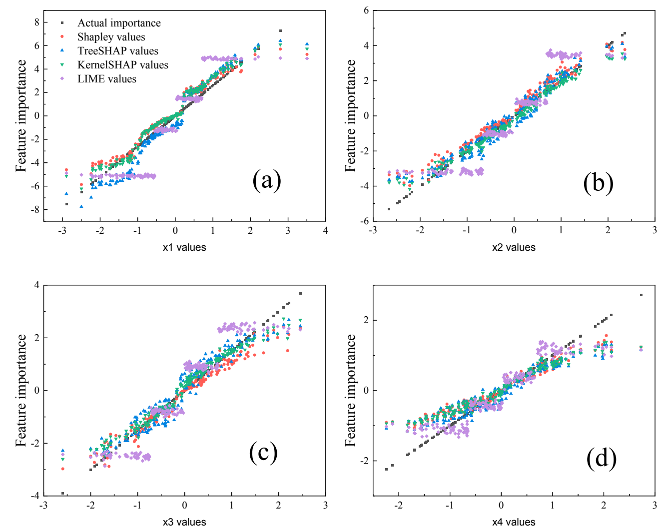


Figure 5. Local explanation results for variables $x_1$ to $x_4$ in the case study on simulated data.

Thus, for local explanations, the Shapley values best reflect the true importance distribution of the variables in terms of explanation accuracy, followed by KernelSHAP, TreeSHAP and LIME.

## 4. SHEAR WAVE VELOCITY PREDICTION

### *4.1 Data analysis*

Shear wave velocity is the basis for calculation of rock mechanical parameters and plays a crucial role in well logging compressibility evaluation. Array acoustic logging is a common method that can provide compressional and shear wave velocities of rocks. However, the cost of this logging method is high, so it is not commonly used in large-scale exploration projects. In the absence of array sonic logging data, engineers and geologists need to rely on more common logging data to estimate rock shear wave velocities.

The data for Case Study 2 are from the Songliao Basin in China, where a suite of dark-colored mud shales is extensively developed in the Qing1 member of the Upper Cretaceous Qingshankou Formation, with active hydrocarbon shows and some exploration prospects. Conventional logging data from well A, acoustic time difference (AC, $\mu s \cdot m^{-1}$), neutron porosity (CNL, %), bulk density (DEN, $g \cdot cm^{-3}$), gamma ray (GR, API), and deep lateral resistivity (RD, $\Omega \cdot m$), were used as inputs with shear wave time difference (DTS, $\mu s \cdot m^{-1}$) as output (Fig. 6).

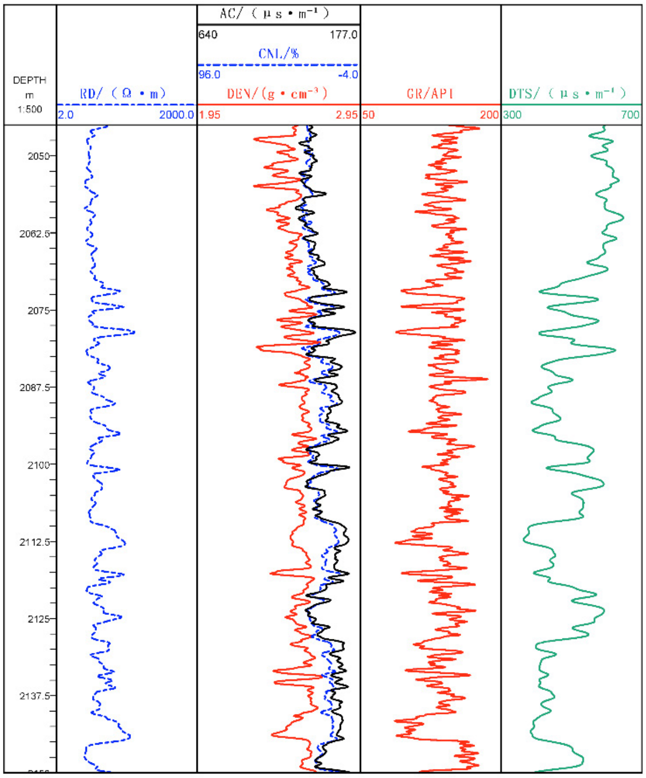


Figure 6. Logging data for well A.

*4.2 RF for regression*

Similarly, the data were randomly divided into training and test sets according to the 8:2 principle, and a random forest model was used as the fitting model. The model hyperparameters are the same as those used in *3.2*.

Fig. 7 shows the prediction results of the random forest model on the test set. The model is highly accurate with a root mean square error (RMSE) of 20.35$\mu s \cdot m^{-1}$ and a coefficient of determination ($R^2$) of 0.98.

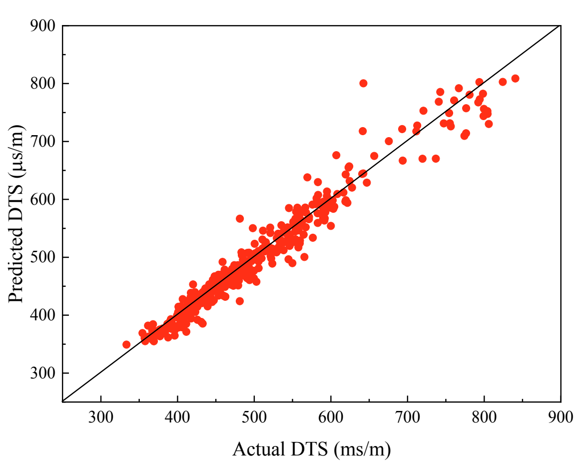


Figure 7. RF model prediction results

*4.3 Explanation methods comparison*

Fig. 8 shows the importance ranking of the four interpretation methods for the logging parameters with similar results. All four methods, Full Shapley, TreeSHAP, KernelSHAP and LIME, recognize the first three important parameters AC > CNL > RD. DEN and GR are similarly ranked in terms of importance, both being lower. It means that all four explanatory methods can provide accurate results in ranking the importance of variables in the shear wave velocity prediction problem.

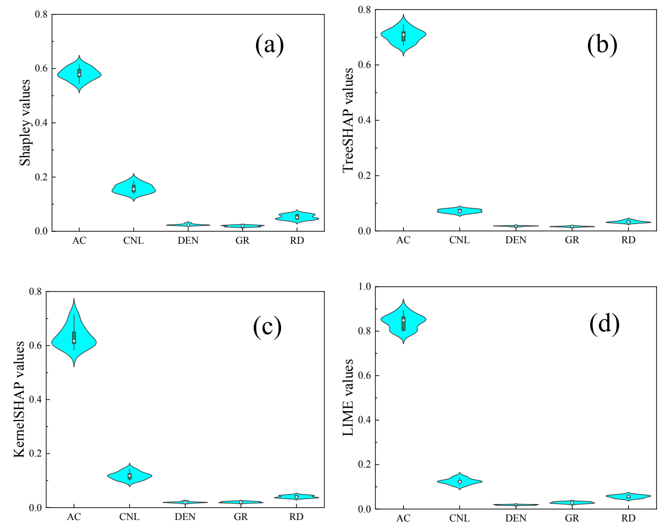


Figure 8. Ranking of variables importance. (a) Full Shapley, (b) TreeSHAP, (c) KernelSHAP and (d) LIME.

Unlike the simulation example in Section 3, the matrix coefficients can be considered as the true importance values of the variables. In practical applications, the Full Shapley value is usually chosen as the actual importance, since the true value of the variable is not known. The RI values between the interpretation results of TreeSHAP, KernelSHAP and LIME methods and Full Shapley values are calculated, and the results of 20 runs of experiments are shown in Fig. 9. Similar results to Fig. 3 can be obtained with the global interpretation accuracy ranked as KernelSHAP > LIME > TreeSHAP.

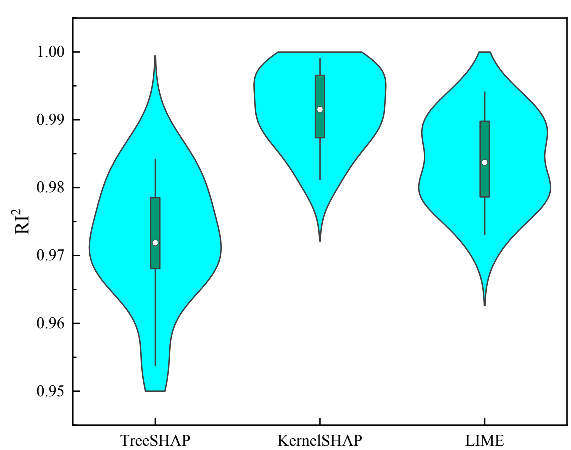


Figure 9. $RI^2$ distributions for 20 trials of the three interpretation methods.

Further, taking the local interpretation of AC as an example, plots of the four interpretation results as a function of AC are produced (Fig. 10). The black, red, blue and green dots represent the Full Shapley value, the TreeSHAP value, the

KernelSHAP value and the LIME value, respectively. It can be seen that the KernelSHAP value is closest to the Full Shapley value, followed by the TreeSHAP value. The least effective is the LIME method, which exhibits the characteristics of clustering. Therefore, a similar conclusion to that in 3.3 can be obtained as well, with the local interpretation method accuracy ranked as KernelSHAP > TreeSHAP > LIME.

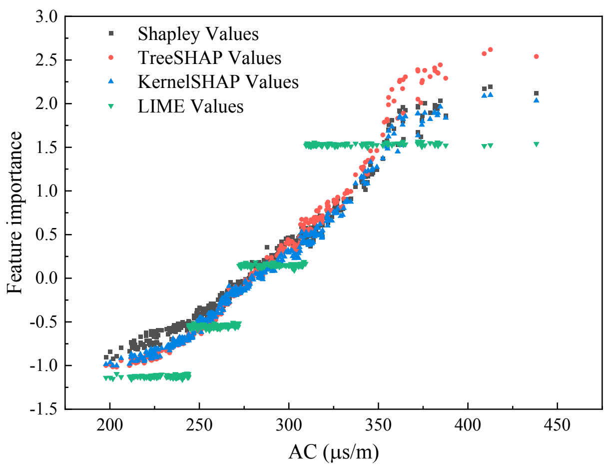


Figure 10. Local explanation results for the acoustic time difference (AC).

## 5. CONCLUSIONS

This study compares and analyzes the performance of Full Shapley, TreeSHAP, KernelSHAP and LIME interpretation methods in logging evaluation problems as modelled with a random forest. The following conclusions are drawn:

- All four methods provided reliable interpretive results.
- For global interpretation, the order of interpretation accuracy for the four methods is Full Shapley > KernelSHAP > LIME > TreeSHAP.
- For local interpretation, the order of interpretation accuracy of the four methods is Full Shapley > KernelSHAP > TreeSHAP > LIME.
- It is worth mentioning that the data used in this study are all low-dimensional, so there is a problem of computation time that can be ignored. However, in future studies, when targeting high-dimensional data, use of TreeSHAP and LIME are recommended. Although these explanatory models could be marginally less reliable than Kernel SHAP or Full Shapley models, they have a faster computation speed.

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